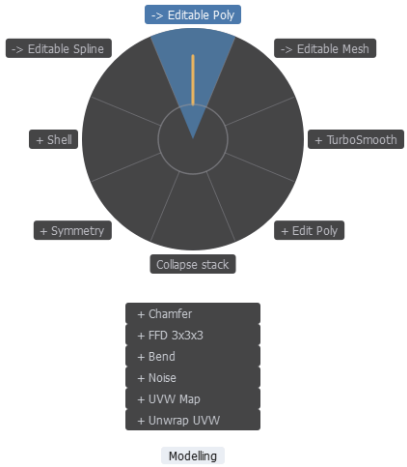
MarkingForge

QuickStart - the short version

3DS MAX 2027 · VERSION 1.5.94 · ABOUT 15 MINUTES



What you need to start: 16 short lessons in four parts, then the tips and tricks. Everything else is in the **QuickStart Long** and the **Reference** - **MarkingForge > Documentation** opens them.

New since 1.5.70 - read this first

What changed how MarkingForge feels - each takes a second to try. Questions and ideas: the MarkingForge Discord, **https://discord.gg/HsakZWQZAw**.

You never have to hit a tile to confirm

A dial picks by the **direction** you move in, not by the caption under the cursor. Move towards a command and let go anywhere that way - near the centre, on its caption or far past it: the command runs. Only the **centre** cancels. In Maya's shape the caption under the cursor wins.

The dial opens in a tenth of a second (1.5.72)

Hold the key with the mouse still: the dial is there after 100 ms - 3ds Max alone waits 250 ms. **Dial speed...** in the editor sets 0 to 250 ms and an opening animation. A flick - moving at once - never waits: the command runs and no dial is drawn.

Your size (1.5.70, 1.5.73)

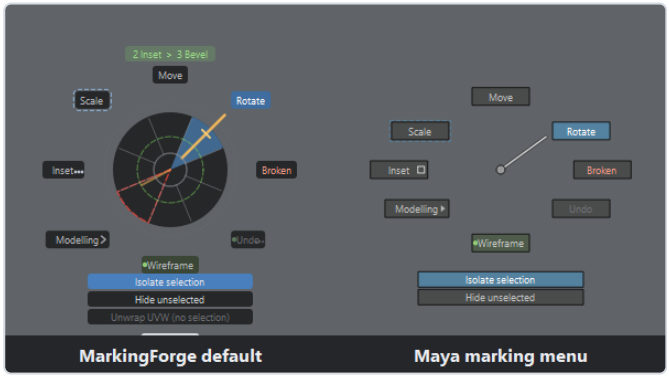
Dial size... in the editor: the whole dial 60 to 160 %, the text 70 to 200 %, and the green return ring of multiple picks - with a picture of the real dial while you drag.

Any key - Tab too (1.5.73, 1.5.75)

A dial can sit on any key. On Tab it opens from the viewport and the panels alike and never walks the keyboard into a number field. Ctrl+Z silent while the toolbar's Undo works? The keyboard is in a field: press Esc.

Exactly Maya's marking menu (1.5.72 - 1.5.74)

One tick: **MarkingForge > Experimental features > Maya marking menu look**. Every dial is then drawn as in Maya: no disc, grey boxes round a small centre dot, a line to the cursor, Maya's blue on the box you aim at, and Maya's **option boxes** - let go on the small square beside a command to open its settings. Clear the tick and your own colours are back. For one dial only: **Colours...** > preset *Maya marking menu*, or just its **Shape** list.



The same dial in both shapes: the MarkingForge dial (left) and the Maya marking menu look (right) - Maya's option box beside *Inset*.

Newest (1.5.80 - 1.5.85)

A tap repeats your last modelling command (Experimental features > Tap repeats the last modelling command - poly, mesh and spline each on their own). **Nothing is lit at rest**, a quick tap runs nothing. **Right-click cancels** a held dial. *No object outline in sub-object mode* hides the object behind.

Before you start

1. Look at the 3ds Max menu bar. You should see a **MarkingForge** menu.
2. Open **MarkingForge > Set up standard shortcuts...** It gives each dial its key, or says they already have one.

A dial is a round menu under a key

Hold Alt+1: eight commands appear around the mouse.

Aim, do not read

Move towards a command and let go of the key. Your hand learns the direction.

Nothing can go wrong

Let go in the middle, or press Esc: nothing runs. Ctrl+Z undoes any command.

About the keys: a safe start, meant to be changed

The dials come on **Alt+1..8**, **Ctrl+Alt+1..8** and **Shift+Alt+1..8**. These keys were chosen because they stay out of the way of 3ds Max's own shortcuts - MarkingForge takes none of them, and the setup skips any key you already use.

They are a starting point, not a rule. Put the dials you use most on keys your hand already knows - Space, a letter, a mouse button:

1. Open **MarkingForge > Menu editor...** and click the dial in the list.
2. Click the field **This slot's shortcut** and press the new key. The editor warns when 3ds Max or another dial already uses it.
3. Press **Apply shortcuts**. **Undo** next to it brings the previous keys back. (**Mouse button**, a little lower, opens the dial with a mouse button instead.)

This QuickStart names the standard keys - with your own, use yours.

PART 1

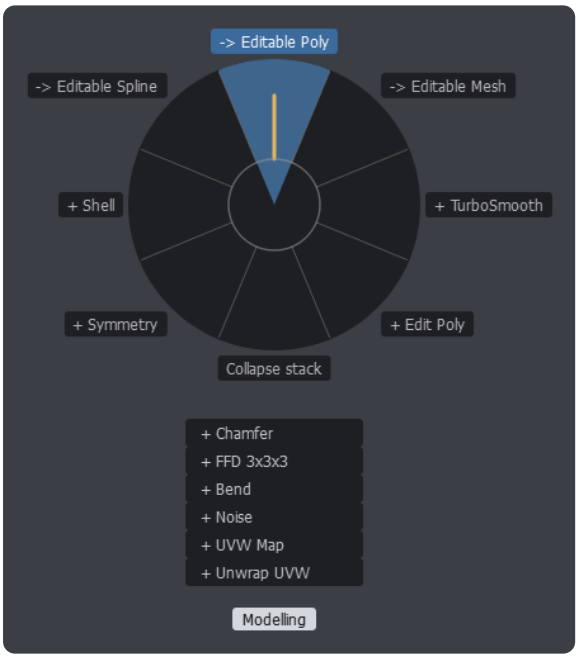
Using the dials

Five lessons. Open an empty scene and try each step.

1 Your first gesture

1 min

1. Mouse over a viewport. Hold **Alt+4** - the Create dial.
2. Move up to **Box** and let go. Drag a box in the viewport.
3. Click the box. Hold **Alt+1**, move up to **-> Editable Poly**, let go.



Alt+1, moved up: the box becomes an Editable Poly.

RESULT

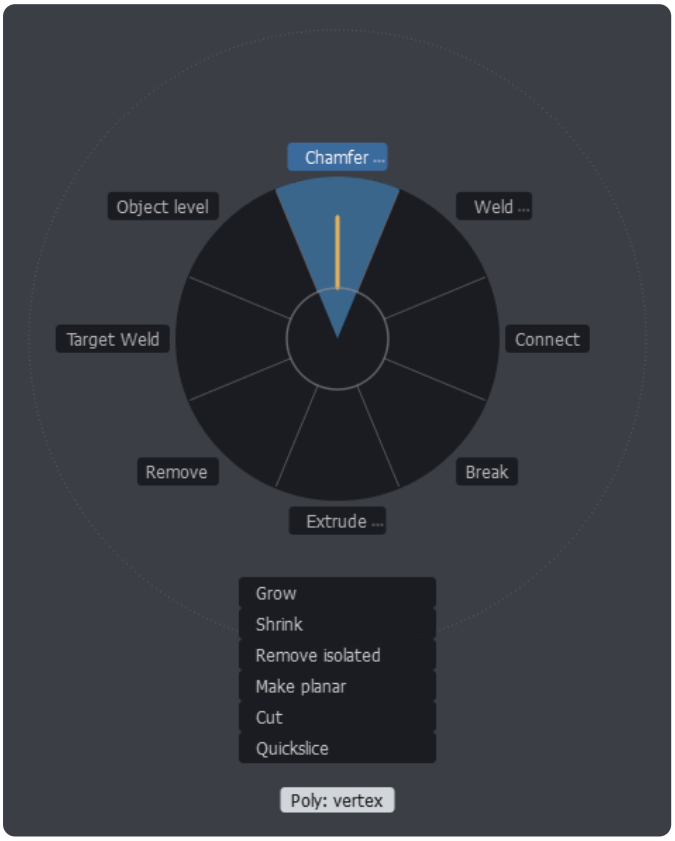
Two gestures, no menus.

You do not have to hit the caption. The direction counts: release anywhere towards the command - near the centre or far past its caption - and it runs. Only the centre cancels.

2 The dial changes with your work

1 min

1. Press **1** (vertices). Hold **Alt+1**: now it shows vertex tools - Chamfer, Weld, Connect...
2. Press **4** (polygons). Hold **Alt+1**: Inset, Extrude, Bevel...
3. Move up-left (**Object level**) to go back to the whole object.



Vertices.



Polygons.

3 Flick - the fast way

1 min

Press the key and move **at the same moment**, then let go. The command runs and the dial does not even show. Try each one a few times:

Press, then move	What happens
Ctrl+Alt+7, up	Zoom to see everything
Alt+3, up / down	Select all / select nothing
Ctrl+Alt+1, up	Move tool
Alt+7, up	Undo one step

4 Near or far: the dotted circle

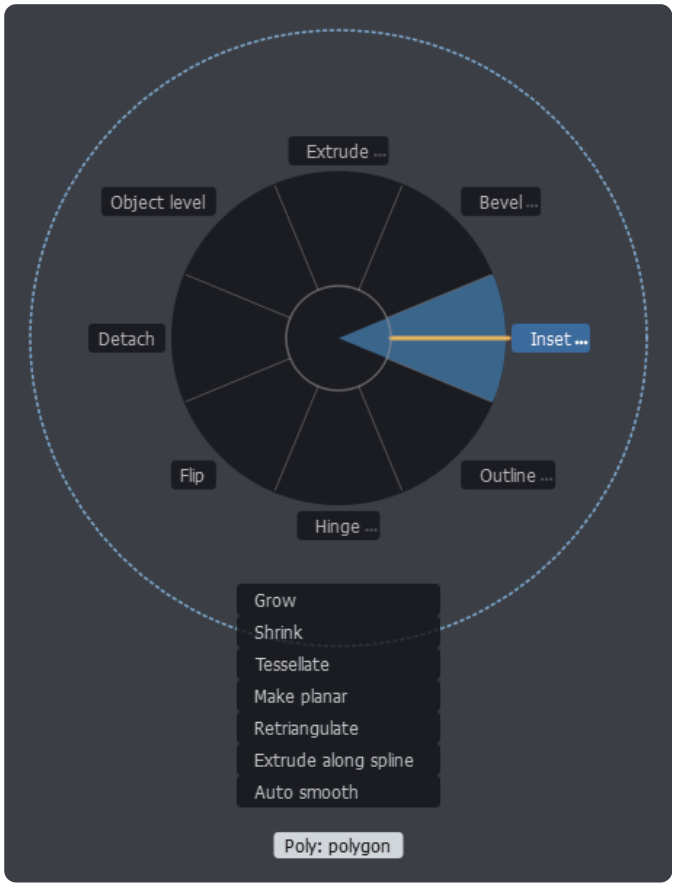
½ min

Commands with three dots (Chamfer..., Inset...) have a settings window.

- Let go **near** the dial: the command starts at once - drag in the viewport to set it.
- Let go **past the dotted circle** (it lights up): the settings window opens.



Near: Inset starts at once.



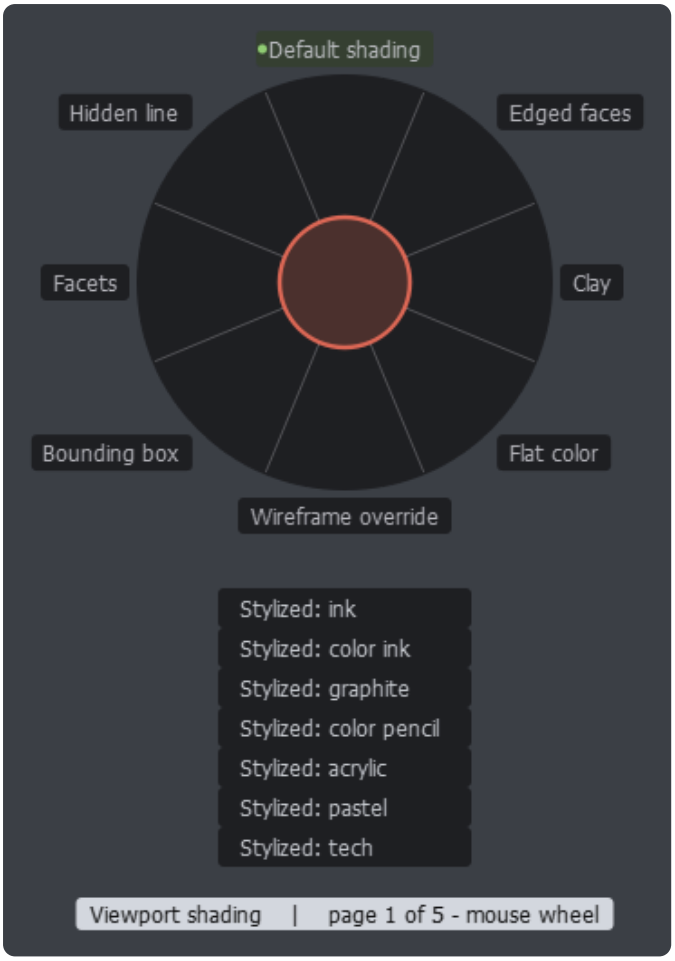
Past the circle: Inset's window.

> **Never want these windows?** Tick **MarkingForge > Experimental features > No settings windows (Bevel, Extrude...)** - no dotted circle, no three dots, every command starts at once.

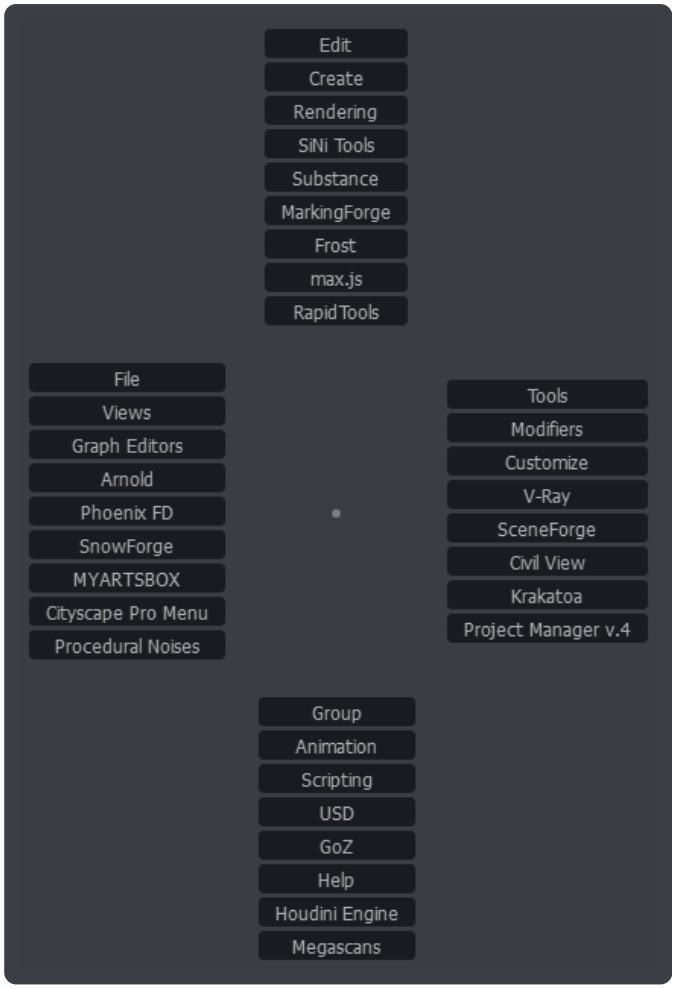
5 Pages, the hotbox and undo

1 min

1. Hold **Alt+5**: how the viewport looks. Keep holding and **turn the mouse wheel** - more pages (lights, what is shown, views).
2. Turn the wheel one step the other way: the **hotbox** - every 3ds Max menu as tiles. Esc closes it.
3. Hold **Alt+7**: your last steps by name. Pick one to undo back to it.



Alt+5 - the viewport.



Alt+5, wheel - the hotbox.

Which key is which

The number (1-8) is the place; **Alt**, **Ctrl+Alt** or **Shift+Alt** is the row.

Key	Dial	Key	Dial	Key	Dial
Alt+1	Modelling	**Ctrl+Alt+1**	Transform	**Shift+Alt+1**	Viewport display
Alt+2	Modifiers	**Ctrl+Alt+2**	Align and pivot	**Shift+Alt+2**	Selection sets
Alt+3	Selection	**Ctrl+Alt+3**	Lights and cameras	**Shift+Alt+3**	Materials
Alt+4	Create	**Ctrl+Alt+4**	UV mapping	**Shift+Alt+4**	Render
Alt+5	Viewport shading	**Ctrl+Alt+5**	Snaps and grid	**Shift+Alt+5**	Animation
Alt+6	Modifier stack	**Ctrl+Alt+6**	Show and hide	**Shift+Alt+6**	Scene
Alt+7	Undo	**Ctrl+Alt+7**	View	**Shift+Alt+7**	Tools and setup
Alt+8	Recent commands	**Ctrl+Alt+8**	Viewports	**Shift+Alt+8**	File

> Print your own list: **MarkingForge > Dial cheat sheet (HTML + PDF)...** - keep it next to the keyboard for the first week.

PART 2

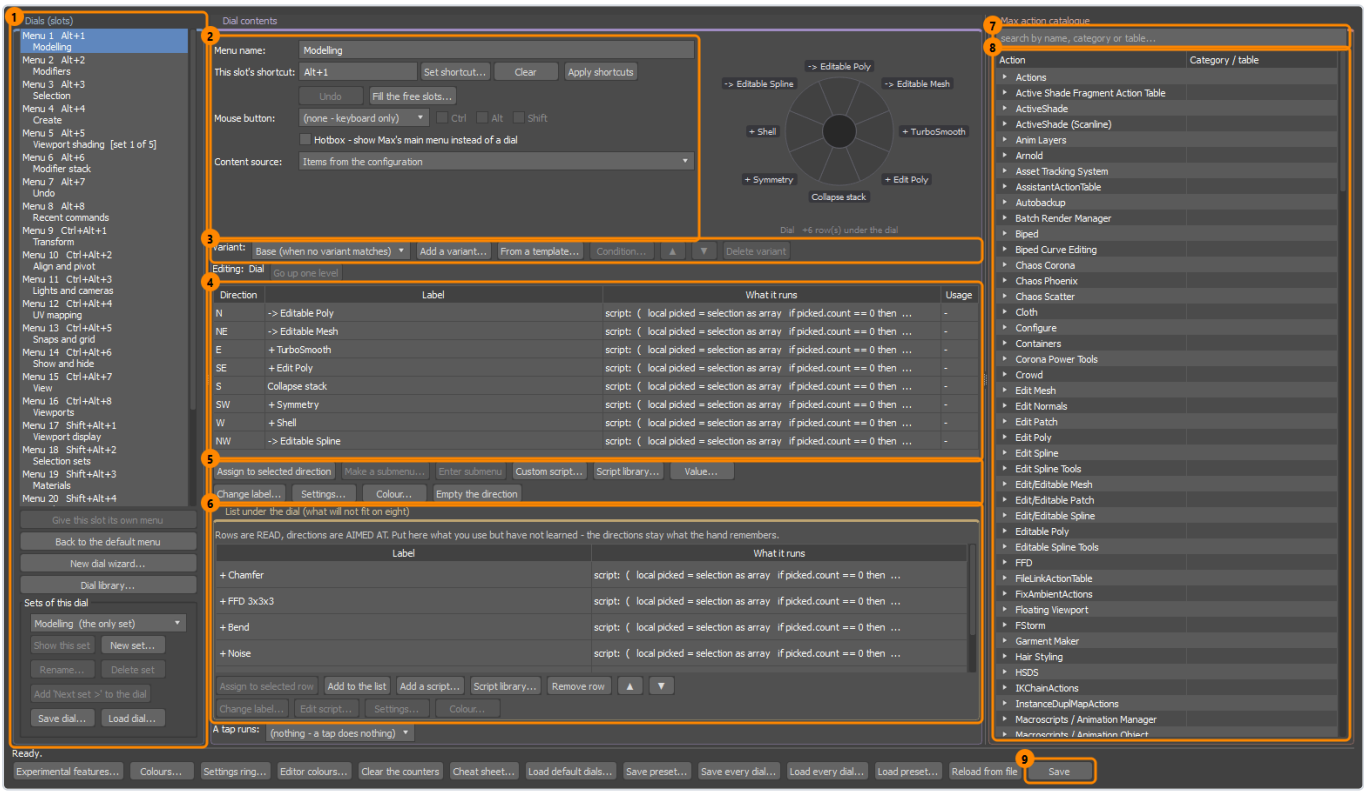
Changing the dials

Two lessons. Nothing changes until you press **Save**.

6 Open the editor

½ min

MarkingForge > Menu editor...



The menu editor.

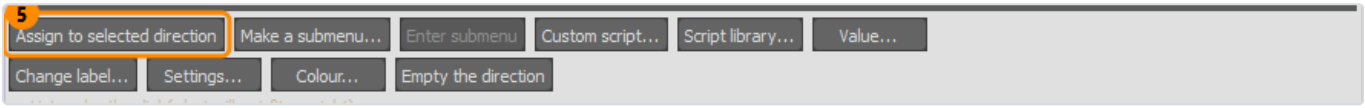
1. The 24 dials - click one to change it.
2. Its name, key and mouse button.
3. Variants - versions of the dial for other situations (lessons 8-11).
4. The eight directions.
5. Buttons for the selected direction.
6. Extra rows under the dial.
7. Search box.
8. Every 3ds Max command.
9. Save.

7 Put your own command on a dial

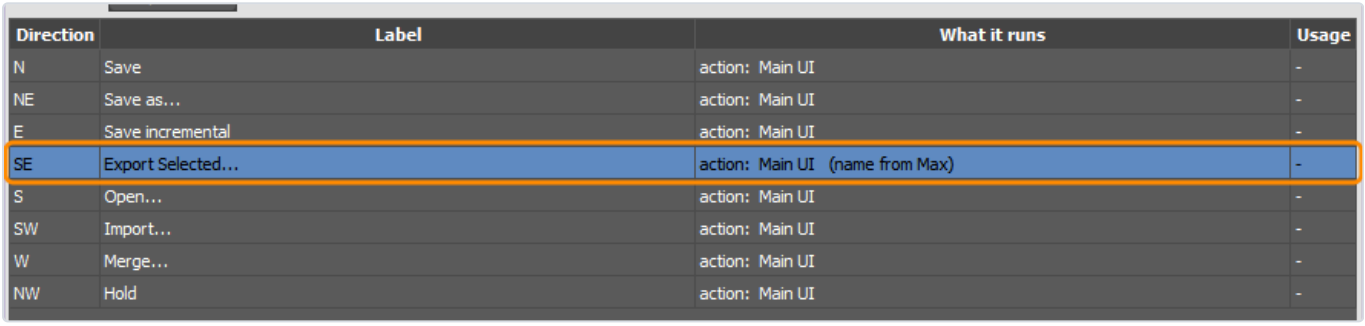
1 min

Example: on the File dial (Shift+Alt+8), put **Export Selected** down-right.

1. Click **Menu 24** in the list of dials.
2. Type `Export Selected` in the search box and click the command.
3. Click the **SE** row in the directions table.
4. Press **Assign to selected direction** under the table (or double-click the command).
5. Press **Save**.



The button under the directions table.



Direction	Label	What it runs	Usage
N	Save	action: Main UI	-
NE	Save as...	action: Main UI	-
E	Save incremental	action: Main UI	-
SE	Export Selected...	action: Main UI (name from Max)	-
S	Open...	action: Main UI	-
SW	Import...	action: Main UI	-
W	Merge...	action: Main UI	-
NW	Hold	action: Main UI	-

SE now runs Export Selected.

> **TRY IT**
>
> Hold Shift+Alt+8 over a viewport: down-right is Export Selected.

PART 3

Variants: one key, the right dial for each situation

Four lessons. A variant is the same dial in another situation - another object, another modifier, another window.

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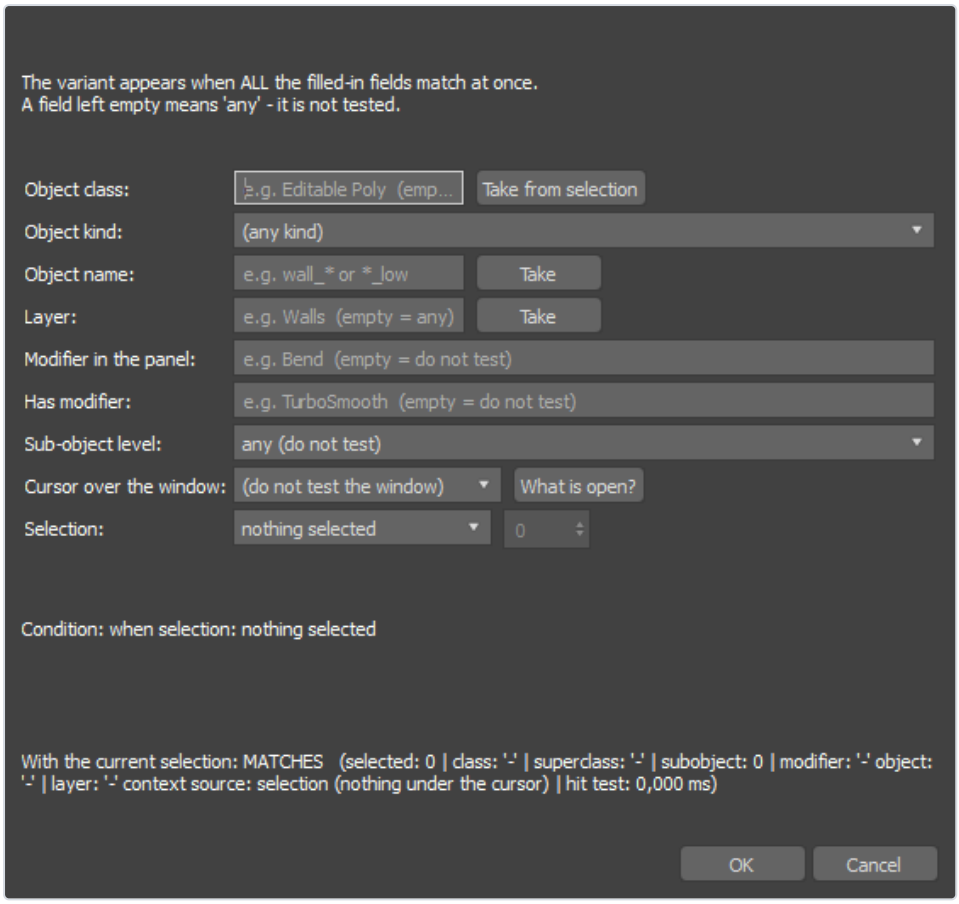
8 Variants: the dial changes by itself

1½ min

In lesson 2 Alt+1 showed other tools for vertices and for polygons. Those are **variants**: other versions of the same dial, each with a **condition**. When you press the key, the dial shows the first variant whose condition is true - and its normal contents, the **base**, when none is.

Example: after lesson 7, down-right on the File dial is **Export Selected**. With nothing selected it has nothing to export - so let that direction export the whole scene then.

1. Click **Menu 24** in the list of dials.
2. Press **Add a variant...** - in the *Variant* row above the directions table.
3. In the window that opens, set **Selection** to *nothing selected* and press **OK**. The new variant starts as a copy of the dial, and the *Variant* list now shows it.
4. Type `Export` in the search box, click **Export...**, click the **SE** row and press **Assign to selected direction**.
5. Press **Save**.



Selection set to *nothing selected*. Fill in only what must be true - an empty field is not tested.

TRY IT

Click empty space in a viewport so nothing is selected and hold Shift+Alt+8: down-right is Export. Select an object: it is Export Selected again.

A variant can appear when...	For example
the object is of a class or kind	Editable Poly, any light
a modifier is open in the Modify panel	Unwrap UVW, Edit Poly
you are on a sub-object level	vertices, polygons
the cursor is over a window	the Slate material editor, the UV editor
the object's name or layer matches	wall_*, the layer Walls
this many objects are selected	nothing, one, two or more

To change the base again, pick **Base** in the *Variant* list. The first variant that matches wins, so a narrow one goes above a wide one - ▲ and ▼ move it.

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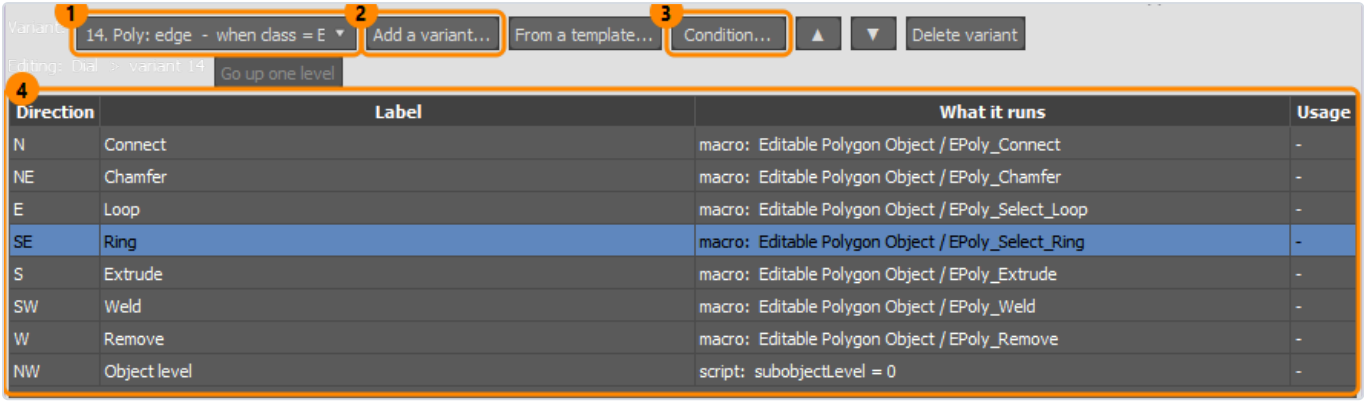
9 The variants that are already there

½ min

The built-in dials are full of variants - reading them is the quickest way to learn how they work.

1. Click **Menu 1** and open the *Variant* list above the directions table.

2. Pick a few and look at the directions table - and at the condition next to the name.



The Variant list (1), Add a variant (2), Condition... (3) and the directions of the variant picked (4).

Variant of Menu 1 (Alt+1)	Its condition	What the dial shows
Materials (Slate)	cursor over the Slate material editor	material commands
Track View	cursor over Track View	key tangents, Flatten, Frame all
UV (Unwrap)	modifier in the panel: Unwrap_UVW	Quick relax, Pack, Stitch...
Edit Poly: edge	modifier: Edit_Poly, sub-object level 2	Connect, Chamfer, Loop...
Poly: edge	object class: Editable Poly, level 2	the same edge tools
Spline	object kind: shape	-> Editable Spline, + Extrude, + Lathe...
Nothing selected	selection: nothing selected	Box, Sphere, Select by name...

Why the order matters: an Edit Poly modifier on an Editable Poly object matches both *Edit Poly: edge* and *Poly: edge*. The list is read from the top and the first match wins - so the windows come first, then the modifiers, then the object classes, and *Nothing selected* last.

> **TRY IT**
> Create a box, convert it to Editable Poly, press 2 (edges) and hold Alt+1. Now move the cursor over the Modify panel and hold Alt+1 again: the *Modifier stack* variant - the cursor decides too.

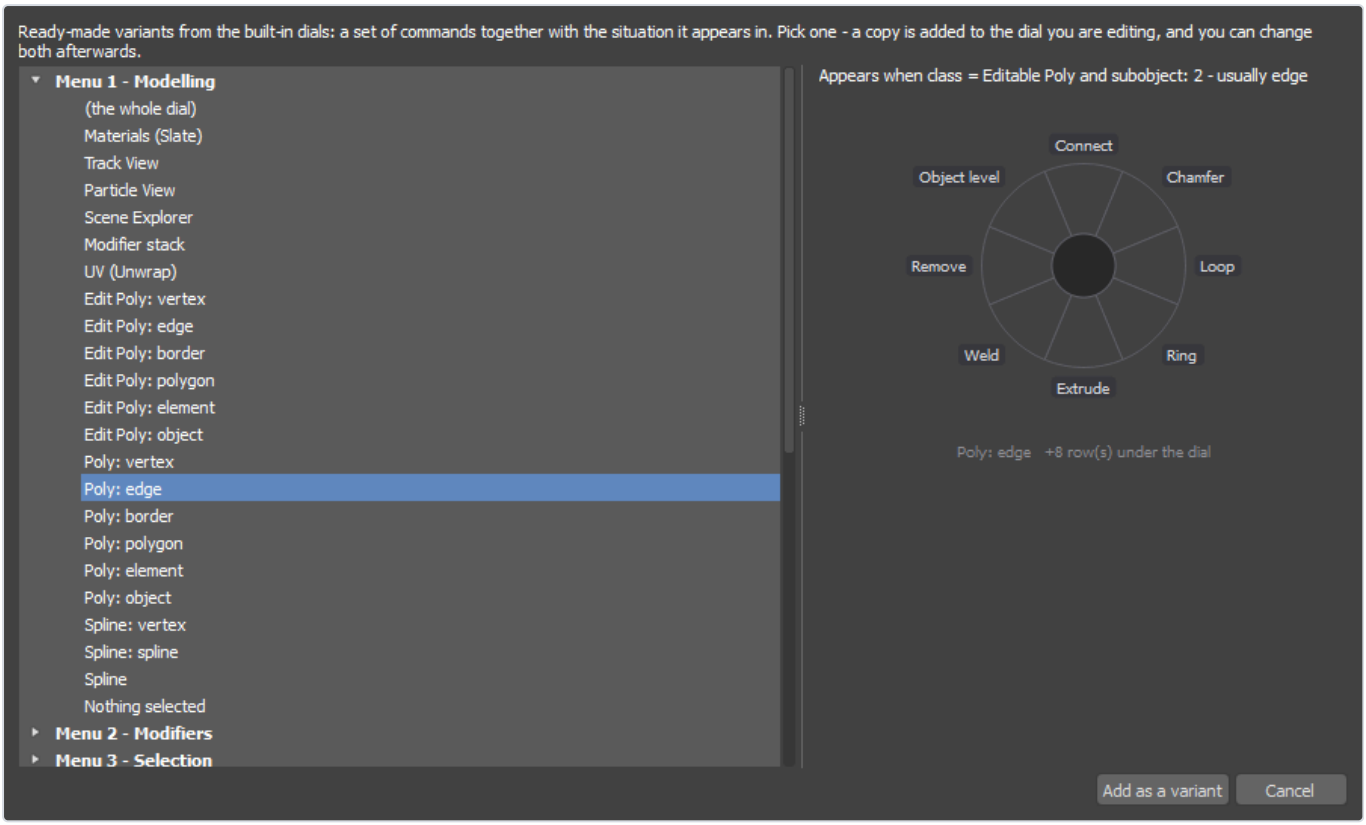
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10 A variant for a window: Track View

½ min

The window under the cursor can be a condition too: the 3D view, the Modify panel, the Slate material editor, the UV editor, Track View, Scene Explorer, Particle View...

1. Open **Graph Editors > Track View - Curve Editor**, select an animated object, hold **Alt+1** over the curves: the Track View variant - tangents, Flatten, Frame all.
2. Now give the Create dial (Alt+4) the same: click **Menu 4** and press **From a template...** (next to *Add a variant...*).
3. Under **Menu 1 - Modelling** pick **Track View**. The right side shows the dial and when it appears. Press **Add as a variant**.
4. Press **Save**.



From a template...: ready-made variants of every built-in dial, with their conditions.

> **TRY IT**
>
> Hold Alt+4 over the curves: the Track View tools. Over a viewport: the Create dial as always.

For your own window condition, use **Cursor over the window** in the condition window. A window that is not in its list? Open it, press **What is open?** next to the list, and pick it there.

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11 Object class, kind and modifier names

1 min

Three fields of the condition window ask about the selected object. They are easy to mix up:

Field	What it looks at	Example
Object class	what the object IS - the bottom of its modifier stack. An Editable Poly with a TurboSmooth on top is still an Editable Poly.	Editable Poly, Editable Mesh, Box, Line
Object kind	a wider sieve: every object of a kind, whatever its class. Picked from a list.	geometry, shape, light, camera, helper
Modifier in the panel	the modifier you have OPEN in the Modify panel right now.	Edit Poly, Unwrap UVW, TurboSmooth, Bend
Has modifier	the modifier is anywhere in the stack, open or not.	TurboSmooth

How to write a name. The name of the TYPE, as the Modifier List and the Create panel show it - not the name you gave an object or a modifier: a modifier renamed to *My unwrap* is still *Unwrap UVW*. A space and an underscore count the same (`Edit Poly` = `Edit_Poly`), and so do capital and small letters. The built-in dials write `Unwrap_UVW` and `Editable Poly` - both work.

Easiest: let the editor write it. Example - a variant of the Modifiers dial (Alt+2) for the Bend modifier:

1. Create a box and add a **Bend** modifier. Keep Bend open (highlighted) in the Modify panel.
2. In the editor click **Menu 2** and press **Add a variant...**
3. Press **Take from selection**. It fills Object class (*Box*), Object kind (*geometry*) and Modifier in the panel (*Bend*) - spelled exactly the way the plugin compares them.
4. Clear **Object class** and set **Object kind** back to *(any kind)*, so the variant works on every object with Bend open - not only on boxes. Press **OK**.
5. Put the commands you use with Bend on its directions and press **Save**.

> The line at the bottom of the condition window says whether the condition *MATCHES* what is selected now - check it before OK. Object name (with `*`, e.g. `wall_*`) and Layer work the same way, with their own **Take** buttons.

PART 4

Settings, chains and your own dial files

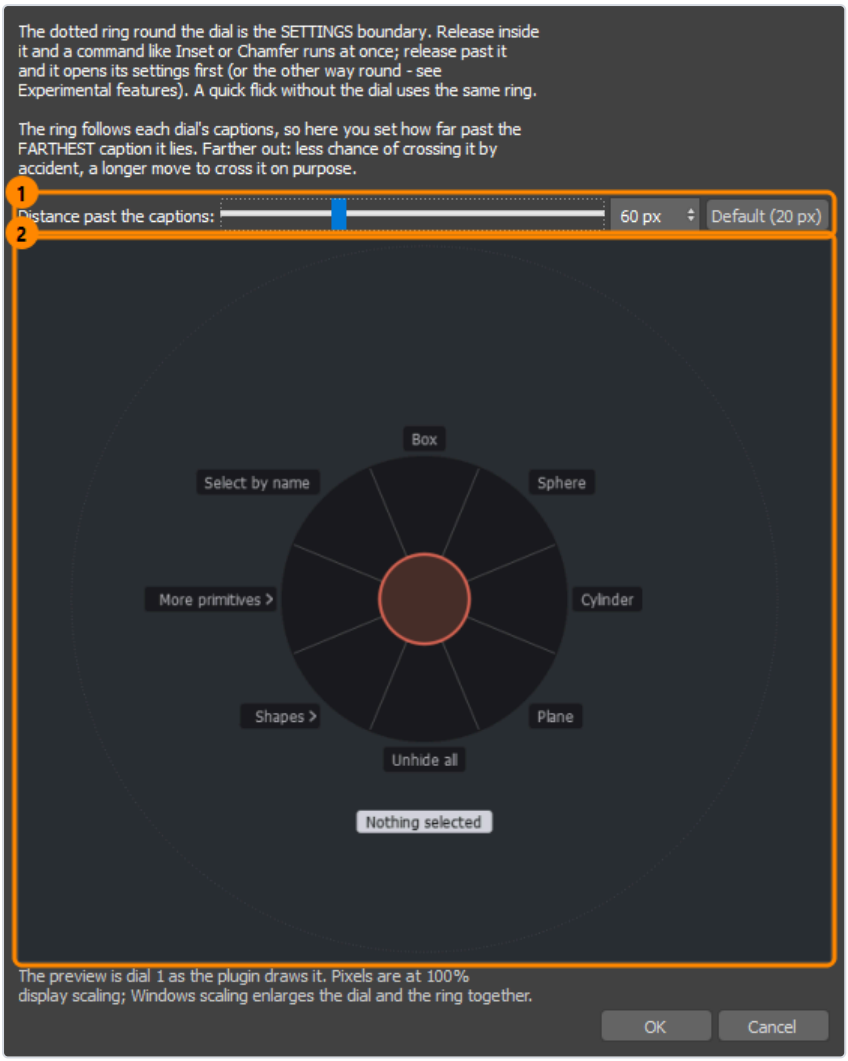
Five lessons.

12 The dotted ring: near or far, and its size

½ min

The dotted ring round the dial (lesson 4) is the **settings** boundary. Let go inside it and a command such as Inset or Chamfer starts at once; let go past it and its settings window opens first. A quick flick without the dial uses the same ring.

1. Press **Settings ring...** at the bottom of the editor.
2. Drag the slider (or type a number): the distance of the ring past the farthest caption, from 0 to 200 px. The picture is the real dial, redrawn as you drag.
3. Press **OK**, then **Save**. **Default (20 px)** puts it back.



Settings ring...: the distance past the captions (1) and the real dial with its ring (2).

- **Farther out**: less chance of opening a settings window by accident - a longer move to open it on purpose.
- **Closer in**: the settings windows are a short move away.
- One setting for every dial. The ring follows each dial, so it always sits just past its captions.
- **The whole dial** bigger or smaller: **Dial size...** beside it, 60 to 160 % (1.5.70).

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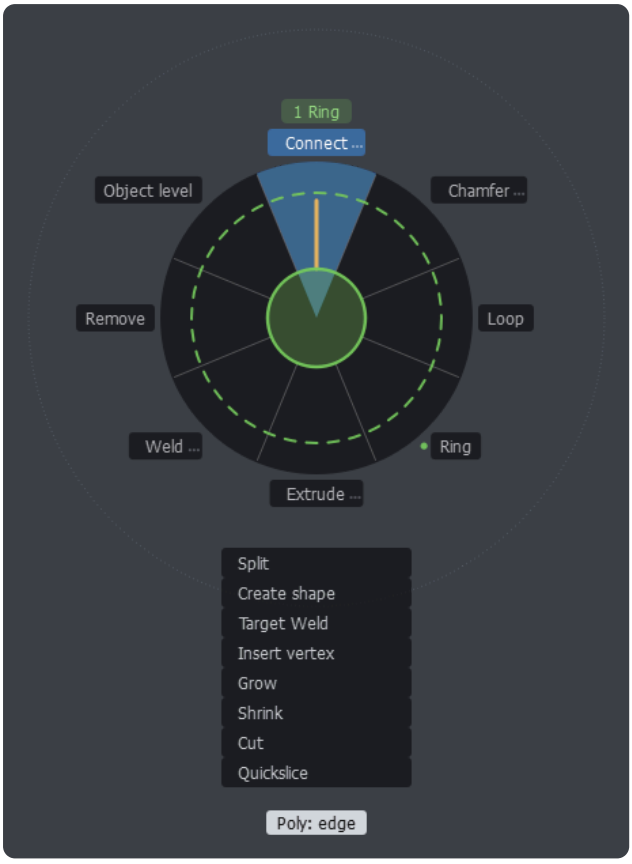
13 Several commands in one gesture

1 min

With one more switch, a single gesture can queue several commands and run them all when you let go - in order, and **one Ctrl+Z** undoes them all. A command that needs your drag (Inset, Bevel...) gets its own turn - lesson 14.

1. In the 3ds Max menu bar open **MarkingForge > Experimental features** and click **Multiple picks in one gesture** (it gets a tick).
2. Hold the key, move out **past the dotted ring** towards a command, and come **back to the dashed circle** near the middle - without letting go. The command waits in a queue: a numbered row above the dial.
3. Queue more the same way.

4. Let go **in the dashed circle** (it turns green): the queue runs, first to last. Or let go **over a command**: the queue runs, then that command.



A command waits in the queue (the row above the dial); the dashed circle round the middle is where you come back to.

Try it - a new edge loop in one gesture:

1. Create a box, convert it to Editable Poly, press **2** (edges) and click one edge.
2. Hold **Alt+1**: out to **Ring** and back, out to **Connect** and back.
3. Let go in the dashed circle. Ring selects the edges round the box, Connect puts a new loop through them. One Ctrl+Z takes both back.

In a queue	Why
Works: commands that finish at once - Loop, Ring, Grow, Shrink, Flip, Connect, Weld, Cap, adding a modifier, display and view commands, your own scripts	Each one is done before the next starts.
Waits for you: commands that need a drag in the viewport - Inset, Extrude, Bevel, Chamfer, Outline	The queue stops at it until you right-click - lesson 14.
Not queued at all: a submenu, a number slider, an Undo-dial step	There is nothing to run - the dial does not add them.

> Changed your mind? Let go on the dot in the very middle, or over an empty direction - the queue is thrown away and nothing runs. A queue you repeat can become one direction: **MarkingForge > Save last chain as an item...**

14 A queue that waits for you: Inset, Extrude, Bevel

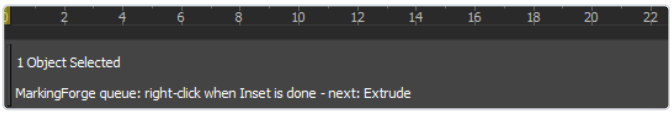
1 min

Inset, Extrude, Bevel, Chamfer and Outline do not finish at once: they switch on a tool and wait for you to drag in the viewport. In a queue each of them gets its turn - the queue stops at it, you do the drag, and a **right-click** moves on to the next command.

1. Switch on **Multiple picks in one gesture** (lesson 13).
2. Create a box, convert it to Editable Poly, press **4** (polygons) and click the top polygon.
3. Hold **Alt+1**: out to **Inset** (east) and back to the dashed circle, then out to **Extrude** (north) and back. The row above the dial reads *1 Inset > 2 Extrude*. Let go in the dashed circle.
4. Inset is on. Drag on the polygon to inset it.
5. **Right-click.** Inset is finished and Extrude starts by itself.
6. Drag up to extrude it, then right-click. Done: a new face inset and pulled out, in one gesture.



Inset (1) and Extrude (2) wait in the queue. Let go in the dashed circle.



While Inset is on, the status line at the bottom of 3ds Max names the next command.

While a tool is on	What happens
Right-click	The tool is finished; the next command in the queue starts. The last one is the command you let go over, if you let go over one.
Esc	Nothing - 3ds Max keeps the tool on. Right-click instead.
Pan, zoom or orbit the view	The queue keeps waiting.

Select, Move, Rotate or Scale (Q, W, E, R)	The tool is finished, as with a right-click - the next command starts.
Another tool (Bevel by hand, another dial command)	The rest of the queue is dropped.
Another object selected, one more, one fewer, or none	The rest of the queue is dropped (the status line says so).
You let go over a value direction (dragged for a number)	The number is written last, after the tool - or dropped with the rest.
A new gesture	The rest of the queue is dropped; the new gesture works as usual.

Ctrl+Z goes back one step at a time here: each command that waited for you is its own undo. Commands that finish at once (lesson 13) still run straight through and undo together. A queue that waited for a tool is not offered by **Save last chain as an item...** - as one item its steps would run straight through.

TRY IT

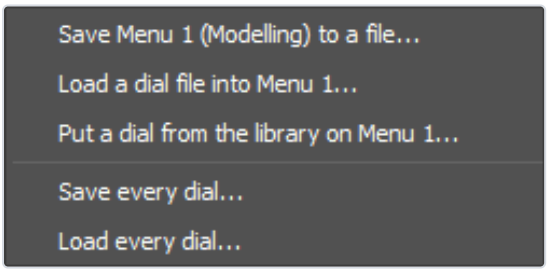
Queue Bevel (north-east) and Inset (east) on a polygon, then let go over Extrude (north): three tools in a row, each finished with a right-click.

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15 Right-click a dial: save it, load it - or every dial

½ min

In the editor, **right-click a dial** in the list of dials (or its picture next to the directions):



The menu of a dial - right-click it in the list.

Command	What it does
Save Menu N (name) to a file...	This one dial, with its variants, in a `.mfdial` file - to keep, or to give a colleague.
Load a dial file into Menu N...	Puts a saved dial on this key. The other dials stay as they are.
Put a dial from the library on Menu N...	The dial library: every built-in, extra and pack dial, ready to take one by one.
Save every dial...	All 24 dials, their pages and their colours, as files in a new dated folder.
Load every dial...	Puts back everything such a folder holds.

TRY IT

Before you change a dial a lot: right-click it and save it to a file. If you do not like the result, load that file back into the same Menu and Save. Nothing else changes.

> Loading changes the editor only - press **Save** to make it work in 3ds Max.

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16 Save, start again, share

½ min

Button (bottom of the editor)	What it does
Save	Your dials work at once.
Reload from file	Throws away what you have not saved.
Load default dials...	The dials MarkingForge came with. Tick *Replace...* to get all of them back.
Save preset... / Load preset...	Your whole layout in one file - to keep or share.

If something goes wrong

Known 3ds Max behaviour: the object behind lights up while you edit sub-objects

Seen in the 2027 version of 3ds Max, possible in other versions: while you edit vertices, edges or polygons of one object, the object **behind** it can light up under the cursor - when you press a dial's key or pan with the middle mouse button, also on an object without modifiers. It shows even during the middle-button pan, which MarkingForge does not touch. A dial works on the object you edit and does not select the one behind (since 1.5.79). If it gets in your way: tick **MarkingForge > Experimental features > No object outline in sub-object mode**, or switch off 3ds Max's *Preview Highlight Outlines Toggle* (Customize > Hotkey Editor, category Views). If the object behind did become selected, Ctrl+Z gives your selection back.

What happens	What to do
Holding the key does nothing	**MarkingForge > Set up standard shortcuts...**
I changed something, but the dial is the same	Press **Save** in the editor.
I want the original dials back	Editor: **Load default dials...**, tick *Replace...*, Save.
Where is the hotbox?	Hold Alt+5 and turn the mouse wheel one step away from you.
A caption is red	That command is not in this 3ds Max. Put another one there.
My variant never shows	Select it and press **Condition...** - the line at the bottom says whether it matches what is selected now. A variant higher in the list may match first.
My modifier variant never shows	The modifier must be OPEN in the Modify panel. Use **Take from selection** instead of typing the name - or **Has modifier** if it does not matter which one is open.
The dial is the same over Track View	The variant needs **Cursor over the window: Track View**, and nothing above it in the list may match first.
The next command of a queue does not start	Finish the tool with a **right-click** - Esc keeps it on. Another tool, another object or a new gesture drops the rest of the queue (lesson 14).

Want more? The QuickStart Long shows scripts, number sliders, colours and more pages on one key; the Reference lists every condition and every window a variant can ask for.

Tips and tricks

2 min

The same tips as chapter 16 of the **Reference** - read them once now, and again after a week with the dials.

Learn by flicking, not by reading

Use the held dial for a few days, then start moving before it appears. Speed comes from never waiting for the drawing.

No settings windows, ever

If Bevel, Extrude or Chamfer windows only get in your way, tick **MarkingForge > Experimental features > No settings windows (Bevel, Extrude...)**. The outer ring and the ... disappear and every command starts at once - drag in the viewport for the amount.

Size the dial to your screen

On a large monitor or with the screen far away, make the dial bigger (**Dial size...** in the editor, up to 160 %); on a small one, smaller. Flicks scale with it.

Coming from Maya?

Colours... > *Maya marking menu* draws the dial as Maya does, and **Dial speed...** at 0-60 ms opens it the moment your hand stops. Flicks never wait either way.

Put opposites opposite

Grow N / Shrink S, Hide S / Unhide N. The hand learns one movement and its reverse - the built-in dials follow this everywhere.

Keep NW as "back"

In every sub-object variant NW returns to object level. Reuse the convention in your own variants.

Eight is the limit, not the target

Put rarely used items in the list under the dial; directions are for what the hand should remember.

Short labels

Change label... with two words beats a cut-off 3ds Max name.

Colour the dial, not the item

A different dial colour per slot tells you which dial opened before you read a word. Colour single items only to warn - a red caption on a destructive command.

Specific variants first

Edit Poly at Vertex above Edit Poly above Editable Poly. When a variant never shows, a more general one above it is catching it first.

Let the context do the work

One key that changes with the sub-object level beats five keys. Look at the built-in Modelling dial before building your own.

Window variants

The same key can do one thing in the viewport and another over Slate or Track View - the built-in Materials, Animation and Scene dials show how.

Cross the rim for settings

Need the Chamfer caddy once? Move past the ring. Need it every time? Settings... on that item.

Chain several picks

Switch on *Multiple picks* first. Out past the ring and back to the dashed circle queues an item; release there runs the queue in order as one undo, the dot in the middle cancels. Chain commands that finish at once (Ring, Loop, Flip, Connect, Weld, a modifier). One that waits for a drag (Inset, Extrude, Bevel, Chamfer; on splines Fillet, Chamfer, Outline - also under the Edit Poly, Edit Spline and Edit Mesh modifiers) stops the queue: drag, right-click, and the next queued command starts (since 1.5.44; the status line names it from 1.5.45). While it waits: Select, Move, Rotate or Scale ends the tool like a right-click, Esc does not end it (3ds Max keeps it on), a pan or zoom keeps the queue, another tool, a change in the selected objects (another, one more, one fewer), a new gesture or a new scene drops the rest - the status line says so; each such step is its own undo. A value direction you let go over is the queue's last step: written after the tool, or dropped with the rest (1.5.50). The event log shows QUEUE lines for waiting, resuming and dropping. Save a chain you repeat as a single item with **Save last chain as an item...** (one that waited for a tool is not offered).

Repeat your last modelling command with a tap

Tick **Experimental features > Tap repeats the last modelling command**: a **tap** of a dial's key - press and let go without moving - runs your last poly, mesh or spline command again, the one for the kind of object you are editing (with several editable modifiers on the stack, the topmost one), the same way it ran - with no settings window while *No settings windows* is ticked (1.5.85, 1.5.87). Extrude, Extrude, Extrude: aim once, then tap. A tap item you gave the dial still wins.

Let the dial suggest the next step

With pair learning on (Experimental features, off by default since 1.5.83), MarkingForge notices what you run after what - Inset, then Extrude. Once a pair has repeated twice, the next command is the dial's outlined tile and a **tap** of the key runs it without aiming. Nothing on the dial moves; the memory lasts for the session, and **Diagnostics > Configuration status** shows how often the suggestions were taken.

Point instead of selecting

With *Menu under the cursor* the dial answers for the object you point at and the command runs on it - no click to select first. Pointing at one of several selected objects keeps the whole selection.

Type when you cannot find it

Switch on *Type-to-search*, hold any Alt or Shift+Alt dial and type a few letters: the dial becomes a search over every 3ds Max command. Enter runs the outlined best match.

Undo dial for big steps

Alt+7, aim NE or E: undo two or three steps in one move, by name.

Recent commands for repetition

Alt+8 repeats what you just did, newest first.

Selection sets under the hand

Shift+Alt+2 shows the scene's named selection sets as directions.

The viewport under one key

Alt+5 sets how the viewport is drawn; the mouse wheel turns to lighting, display and views - every switch on a direction or in the list under the dial.

Hotbox for the rare command

Alt+5 and one wheel notch away shows all of 3ds Max's menus - including other plugins' - without hunting through the menu bar. Ctrl+Shift+middle mouse can open it too, if you bind it.

Scripts on a dial

Custom script... accepts any MAXScript. Snapshot the selection first (`local picked = selection as array`) when the code changes objects - the built-in dials do.

Find the words for a condition

`MarkingForge.contextInfo()` tells you the exact class and modifier names with your object selected.

Print the cheat sheet

Cheat sheet... in the editor makes a PDF of your saved dials to keep beside the keyboard for the first week; it tells you the page count when it is done.

Share a setup safely

Save preset... leaves out paths and usage; Load preset... strips scripts unless you allow them.

One key, several tasks

Give a dial sets - *Modelling*, *UV* - and turn the mouse wheel while it is open to reach the others; a key on *Menu N - next set* changes which one the key opens. The same key and directions serve each task in turn.

A whole setup in one folder

Save every dial... before an experiment, **Load every dial...** to come back - every dial, its sets, the colours and every option, exactly as saved. Right-click a dial for the same.

Restore one dial, not all

Dial library... opens on the original of the dial you are editing - **Restore Menu N** is one click.

Groups in two strokes

A submenu holds a group the hand reaches by direction twice - see Create (Alt+4): Shapes S, More primitives SW, Helpers W, Cameras and lights NW.

Back up your dials

Copy `menus.json` - a plain text file - or use **Save every dial...**.

Support, bug reports and feature requests

ISSUES	github.com/Forge-Plugins/MarkingForge/issues
ROADMAP	trello.com/b/7gJVqG8Y/markingforge-features-roadmap
DISCORD	discord.gg/HsakZWQZAw (questions, tips, news - with other users and the author)
E-MAIL	forgeplugins@gmail.com
WEB	https://forgeplugins.tech (forgeplugins.tech - more plugins by the author)

Bug reports and feature requests are best opened as issues on GitHub - the forms there ask for exactly what is needed, and you can follow the fix. Quick questions and tips: the **Discord** channel (**MarkingForge > Discord community** opens it). Anything you would rather not post publicly (licence, purchase) goes to the e-mail address. Every message is read.

Reporting a bug

1. In 3ds Max choose **MarkingForge > Report a bug...** - it copies your 3ds Max version, MarkingForge version, configuration summary and start-up log to the clipboard. It works even when the plugin did not load.
2. Open github.com/Forge-Plugins/MarkingForge/issues and pick **Bug report** - or write an e-mail to forgeplugins@gmail.com instead. Both are links in the window that opens (since 1.5.20), one click each.
3. Say what you did, what you expected and what happened instead, then paste the details with **Ctrl+V**.
4. Add a screenshot if you can. When the problem is in one dial, attach a preset of it (editor > **Save preset...**).

Suggesting a feature

Open an issue with the **Feature request** form, choose **MarkingForge > Suggest a feature...**, or simply write. Describe what you would like to do and how you do it today - a concrete example from your work says more than a list of wishes. Ideas for new built-in dials, commands and options are all welcome. What is already planned and in progress is on the **Roadmap** above - **MarkingForge > Roadmap (Trello)** opens it too.

Nothing is sent by the plugin itself: the details go to the clipboard and your own e-mail program, so you see everything that leaves your computer. MarkingForge has no network access.